

A Quasi-Lumped Ultra-Broadband Contiguous SSL-Diplexer from DC to 80 GHz

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Abstract — A contiguous planar diplexer in suspended stripline technology with ultrawideband channels from DC to 40 GHz and 40 GHz to 80 GHz is presented. The design of the diplexer is based on quasi-complementary LC lowpass and highpass prototypes which are directly transformed into transmission line elements by using a quasi-lumped approach. This very straightforward design procedure requires no susceptance annulling networks and no introduction of unit elements which could limit the diplexer's operation bandwidth. For achieving an optimal compatibility with external circuits, an ultra-broadband SSL-MSL (suspended stripline - microstrip line) transition with a return loss of more than 18 dB from DC to 83 GHz was designed. The realized diplexer is extremely compact (6×4.5mm) and features a return loss of more than 12 dB from DC to 81 GHz with very low channel insertion losses of typically less than 1 dB after proper optimization.

I. INTRODUCTION

Multiplexers are essential parts in RF-frontends where broadband signals have to be splitted into smaller channels for further signal processing. In the past, several authors described multiplexers which showed very good performances but were usually rather narrowband, non-contiguous or both [1],[2],[3]. This is, of course, partially due to the respective applications whose requirements demand only for narrowband designs. But often proposed design approaches would not be suited for broadband contiguous designs because the transmission line networks include redundant unit elements or susceptance annulling networks which generally limit the maximum bandwidth. As pointed out in e.g. [4] or [5], singly-terminated Butterworth prototypes can be used for the design of contiguous diplexers without the need for susceptance annulling networks. To simplify the physical realization, even Kuroda's identities could be used without decreasing the diplexer performance, if the redundant unit elements are applied from the singly-terminated filter's load end only.

In this work, an approach is described which requires neither unit elements nor susceptance annulling networks, thus, enabling the implementation of a contiguous diplexer with channels from DC to 40 GHz and 40 to 80 GHz. This ultra-broadband performance could be achieved by directly transforming the singly-terminated LC-prototypes into quasi-lumped transmission line filters.

Recently, in particular Menzel et al. proposed several high performance quasi-lumped SSL-filters, operating in the fre-

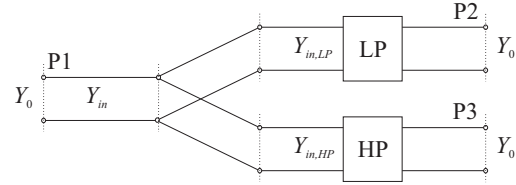


Fig. 1. Parallel-connected diplexer

quency range up to 20 GHz [6],[7]. In [8] several possibilities for realizing quasi-lumped series and shunt reactances have been proposed and even the design of a non-contiguous diplexer, based on doubly-terminated filters, has been shown. For realization, flexible soft-board substrates, such as Rogers 4003 or RT Duroid have been used, which can be easily clamped into a split-block waveguide.

In this paper now, it will be shown that quasi-lumped SSL filters can be designed and manufactured for considerably higher operation frequencies up to 80 GHz and that for this purpose also mechanically sensitive substrates, like alumina, can be used. Furthermore, the applicability of quasi-lumped singly-terminated filters for the realization of ultra-broadband contiguous diplexers will be demonstrated.

II. LC-PROTOTYPE OF THE CONTIGUOUS DIPLEXER

Fig. 1 shows a contiguous diplexer, realized by shunting a lowpass filter (LP) and a highpass filter (HP) at a common junction. To minimize reflections at the input port P1, the complex input admittances $Y_{in,LP}$ and $Y_{in,HP}$ of the two filters should satisfy at all frequencies the equation

$$Y_{in} = Y_{in,LP} + Y_{in,HP} = Y_0 \quad (1)$$

Assuming a purely real reference admittance Y_0 this condition can be splitted into the following two conditions:

$$\text{Re}(Y_{in,LP}) + \text{Re}(Y_{in,HP}) = Y_0 \quad (2)$$

$$\text{Im}(Y_{in,LP}) + \text{Im}(Y_{in,HP}) = 0 \quad (3)$$

Two filters which satisfy those conditions are commonly referred to as complementary filters. As explained e.g. in [9], filters based on singly-terminated Butterworth prototypes (resistive termination at only one filter end) exhibit such a complementary behaviour. In [10] equations are provided how to modify the filter element values of Chebyshev filters to achieve quasi-complementary filters.

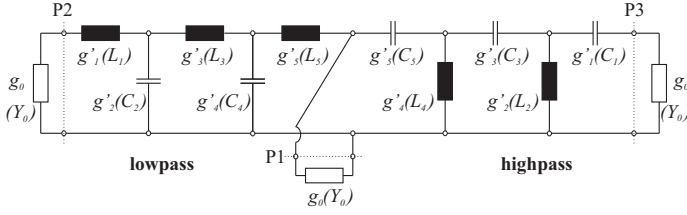


Fig. 2. LC-prototype of a contiguous diplexer with $f_T = 40$ GHz: $L_1 = 179,7$ pH; $C_2 = 120,4$ fF; $L_3 = 382,3$ pH; $C_4 = 138,4$ fF; $L_5 = 324,2$ pH; $C_1 = 88,1$ fF; $L_2 = 131,4$ pH; $C_3 = 41,4$ fF; $L_4 = 114,3$ pH; $C_5 = 48,8$ fF

By means of these equations, a LC-prototype for a contiguous diplexer with a transit frequency f_T of 40 GHz was developed. The lumped element values L_n and C_n of the diplexer, shown in Fig. 2, are based on the modified filter element values g_n of a fifth order singly-terminated Chebyshev filter with 0.5 dB pass-band ripple.

III. QUASI-LUMPED FILTER DESIGN

As explained in the introduction the diplexer should be realized in SSL technique. Thus, as a first step a proper single-mode SSL cross section must be designed. Single-mode waveguides in the millimeterwave frequency range up to 80 GHz require very small cross sections. The dimensions of a WR-12 waveguide (60-90 GHz) e.g. are only 3.1×1.55 mm. The cross section of a SSL with a 127 μ m thick alumina carrier substrate must be chosen even smaller due to the increased dielectric fill factor. In Fig. 3 the necessary dimensions of the SSL channel (determined with a commercial port solver) for single-mode operation range up to about 88 GHz are given.

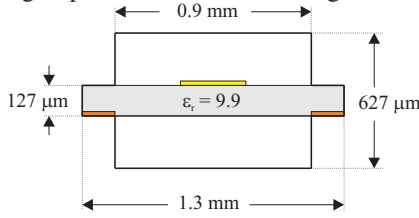


Fig. 3. Cross section of the SSL

This SSL configuration was used for the simulation of the **quasi-complementary filters**. Layouts of the designed lowpass and highpass filters are shown in Fig. 4 (a) and (b). The series inductances of the lowpass were realized by short high impedance lines and the shunt capacitances by short low impedance lines. The total length of the lowpass without feed lines is only 1.5 mm. The series capacitances of the highpass filter were realized by end-coupled line sections which are either located on the same side (C_3) or on opposite sides of the substrate (C_1 and C_5). In case of the realization of the capacitance C_1 , it was necessary to overlap both lines to achieve the required capacitance value of 88.1 fF (in spite of the very thin substrate and the high dielectric constant). The parallel inductances L_2 and L_4 were realized by short-circuited high impedance stubs. The highpass filter is, with a total length of only 0.8 mm, extremely short. For comparison, the effective wavelength of our SSL at 80 GHz is about 2.1 mm.

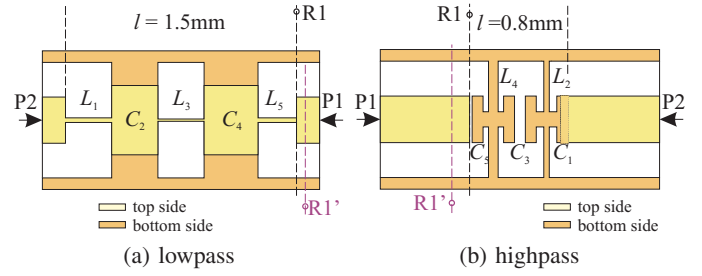


Fig. 4. Quasi-lumped SSL lowpass and highpass filter

In Fig. 5, the simulated complex input admittances of the optimized quasi-lumped filters are compared to the ideal LC-prototypes. The curves are, in spite of the singly-terminated designs, normalized to $Y_0 = 0.02$ S ($Z_0 = 50 \Omega$) at both ports. The transit frequency of the diplexer $f_T=40$ GHz is marked by squares. **Results are analytically de-embedded either to reference plane R1 or R1'.**

The agreement between LC-prototypes and SSL-filters, de-embedded to reference planes R1, is quite good. By adding very short lengths of transmission line at both filters (lowpass: $4^\circ@40$ GHz; highpass: $7^\circ@40$ GHz) the agreement of the complex admittances, especially in vicinity of the diplexer transit frequency, can be even improved (reference planes R1'). It is important for broadband diplexer applications that those line lengths are kept short by all means, to prevent e.g. short-circuiting the highpass by high admittance absolute values in the stopband of the lowpass due to phase progression.

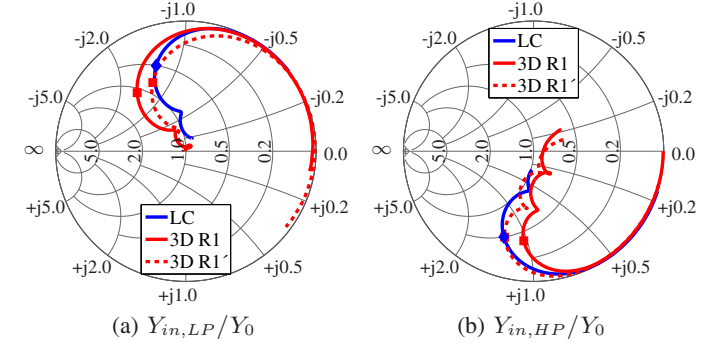


Fig. 5. Simulated complex input admittances of the filters at port P1 (frequency range: DC - 80 GHz)

For a first estimation of the attainable diplexer performance, the single filter input admittances were analytically shunted. This corresponds to connecting both filters at an ideal T-junction in parallel. The resulting S-Parameter of the, in this manner, formed diplexer is shown in Fig. 6 (for reference planes R1'). If such an ideal T-junction could be physically realized, this would result in a contiguous diplexer with a return loss better than 14 dB from DC up to 78.5 GHz.

IV. DESIGN OF THE CONTIGUOUS SSL-DIPLEXER

Based on the design of the two SSL-filters, a contiguous diplexer, according to the LC-prototype in Fig. 2, was developed. In Fig. 7 the layout of the diplexer circuit is shown. Shunting of the lowpass and highpass filter was realized by connecting them directly to the diplexer's SSL input line, **without adding feed lines with significant lengths**. In addition, the input channel and SSL line widths have been slightly

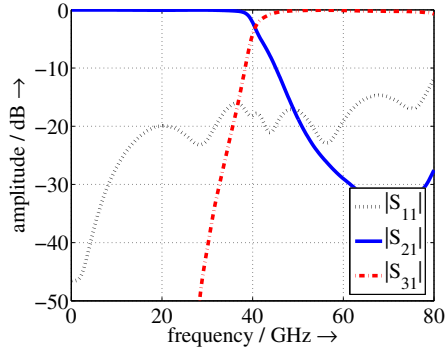


Fig. 6. Performance of a diplexer, formed by shunting analytically the input admittances of the single filters

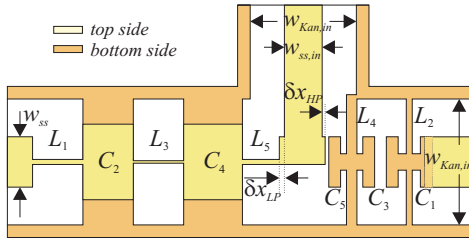


Fig. 7. Layout of the contiguous SSL-diplexer

decreased to further minimize the T-junction dimensions. By these measures hardly any phase shifts can occur, which would affect the broadband operation of the diplexer. For diplexer fine tuning, only electrically very short pieces of transmission lines with the lengths δx_{LP} and δx_{HP} were added at the sides of the input line. Due to the complementary design of the single filters, no further susceptance annulling network is required. Simulation data of the LC-prototype and the 3D-field solver results of the optimized SSL-diplexer are compared in Fig. 8 and Fig. 9. By optimizing the filter elements in vicinity of the T-junction a return loss of more than 20 dB from DC up to 70 GHz could be achieved. This is close to the performance of the ideal LC-prototype. Above 70 GHz the input return loss decreases rapidly to a minimum of 5 dB at 79.5 GHz. The evaluation of the electric field distribution at this frequency showed that the reason for this behaviour is the occurrence of a parallel plate resonance in the T-junction area, due to the enlarged cross-sectional size.

V. REALIZATION AND MEASUREMENT OF THE CONTIGUOUS SSL-DIPLEXER

The realization of such a T-shaped circuit is simple if using soft-board substrates. T-shaping hard-board substrates like alumina (e.g. with a laser) is in comparison rather complicated and expensive. For this reason the SSL-diplexer was splitted into two substrates. Photographs of the realized SSL-diplexer can be found in Fig. 10. For interconnecting the SSLs of the two substrates a small plate, etched out of copper-berrylium sheet was glued to the SSLs. The plate exhibits an enlarged width in the region of the gap between the substrates which was optimized to compensate for the lack of dielectric material. Two 200 μm thick walls, separated by a 700 μm

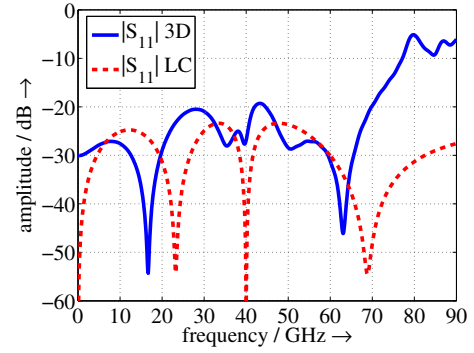


Fig. 8. Comparison of the simulated input return losses of the SSL-diplexer and its LC-prototype

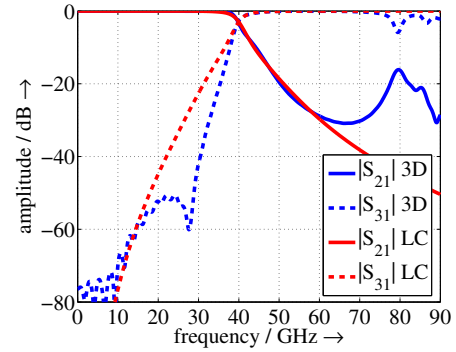


Fig. 9. Comparison of the simulated insertion losses of the SSL-diplexer and its LC-prototype

wide aperture, were provided to simplify the mounting of the feed line substrate (s. Fig. 10). But beside this mechanical advantage those walls also enhance the electrical performance of the diplexer, by essentially reducing the discontinuity caused by the aperture of the SSL input channel. The proposed diplexer requires ground connections (parallel inductances, series capacitances) only at the bottom sides of the substrates. Thus, mounting our pressure-sensitive hard-board substrates is essentially eased because it is sufficient to galvanically contact the substrates only at one side to the waveguide channel. On the other side a 20 μm high gap was provided to account for tolerances of substrate thickness and milling accuracy. Because of the similarities between their electric field distributions, coaxial connectors are typically used for the characterization of SSL-structures. Considering the intended operation

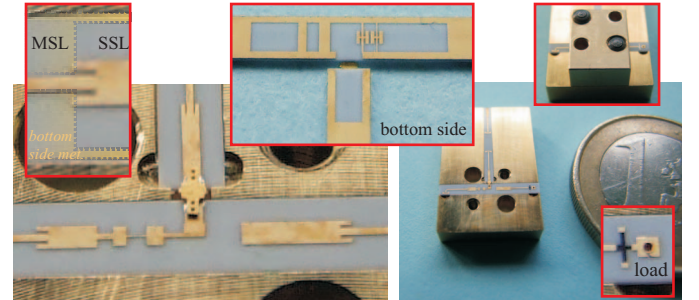


Fig. 10. Photographs of the realized SSL-diplexer

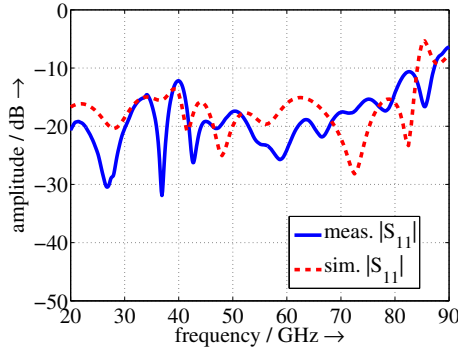


Fig. 11. Simulated and measured return loss of the SSL diplexer

frequency range, this would have required 1mm-connectors in our case which are very expensive. Furthermore, there is the additional drawback that coaxial components are difficult to integrate into larger circuits. Therefore, an ultra-broadband SSL-MSL transition was designed and applied to all three diplexer ports. It consists of a hard SSL-MSL step which is compensated by adding two electrically short fins in front of the SSL. On-wafer measurements of this transition in back-to-back configuration proved an excellent return loss of more than 18 dB from DC up to 83 GHz. Measurements of the single filters by means of this transition, showed a very good agreement with the 3D-simulation data. As our network analyzer provides only two ports, for on-wafer measuring the diplexer (3-port network), two circuits had to be assembled and measured, where the output ports are alternately matched by broadband thin-film resistor loads (s. also [11]).

On-wafer measurement results as well as 3D-simulation results of the realized diplexer circuit can be found in Fig. 11 and Fig. 12. Both, simulation and measurement results, include the SSL-MSL transitions. The first thing one can notice in comparison to Fig. 8 is that the observed return loss peak at the upper passband (parallel plate resonance at T-junction) has been shifted from 79.5 GHz to 85 GHz. This improvement could be achieved by the aforementioned reduction of the input channel aperture at the T-junction through using two substrates.

Measured return loss is better than 12 dB from DC up to 81 GHz. Insertion loss is lower than 1 dB from DC to 35 GHz (S_{21}) and from 52.5 to 74 GHz (S_{31}). Insertion loss at 80 GHz is 1.8 dB. Those values are excellent in respect of the very high frequencies. The agreement between simulation and measurement is very good below 80 GHz, especially if considering the complexity of the circuit. At higher frequencies there are minor deviations, caused probably by manufacturing tolerances and marginal excitation of higher order modes. Size of the entire diplexer circuit, excluding MSL feed lines, is only 6.5×4 mm.

VI. CONCLUSIONS

The design of an ultra-broadband contiguous diplexer in SSL technique with channels from DC to 40 GHz and 40 GHz to 80 GHz has been presented. For realizing complementary low and highpass filters, a straight transformation of the optimal

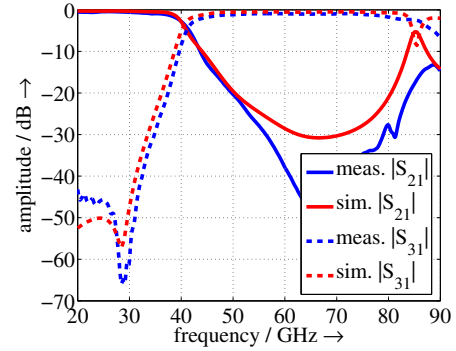


Fig. 12. Simulated and measured insertion losses of the SSL diplexer

LC-prototype into quasi-lumped transmission line structures has been performed. By this approach no susceptance annulling networks are required. For compatibility as well as measuring reasons, an ultra-broadband SSL-MSL transition with a return loss of more than 18 dB (back-to-back) from DC to 83 GHz has been designed. Owing to the quasi-lumped design the contiguous diplexer is with 6.5×4 mm extremely compact and features a return loss of more than 12 dB from DC to 81 GHz with very low channel insertion losses of typically less than 1 dB

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